

ORF 524 — Midterm Exam

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Notes:

- Duration of examination: 180 minutes.
An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
This is otherwise a closed-book and individual examination.

1 Finite-Sample Statistical Inference (50 points)

Let $(X_i : 1 \leq i \leq n)$ be an i.i.d. random sample from $X \sim \mathbb{P}_\theta$, with absolutely continuous d.f.

$$F(x; \theta) = \mathbb{P}_\theta(X \leq x) = \left(1 - \left(\frac{L}{x}\right)^{1/\theta}\right) \mathbb{1}(x > L),$$

where $\theta \in \Theta = (0, 1)$ is an unknown parameter, and $L > 0$ known. All limits are taken as $n \rightarrow \infty$.

1. Find the complete sufficient statistic for θ . Is the statistic minimal?
2. Find the method of moment estimator $\hat{\theta}_{\text{MM}}$ based on $\mathbb{E}_\theta[X_i]$.
3. Show that $\hat{\theta}_{\text{MM}}$ is downward biased but consistent for θ .
4. Find the maximum likelihood estimator $\hat{\theta}_{\text{ML}}$.
5. Show that $\hat{\theta}_{\text{ML}}$ is unbiased and consistent for θ .
6. Find the UMVU estimator of θ . Is the estimator asymptotically efficient?
7. Consider the following one-sided hypothesis testing problem:

$$H_0 : \theta \geq \theta_0 \quad \text{vs.} \quad H_1 : \theta < \theta_0, \quad \text{for some fixed } \theta_0 \in (0, 1).$$

- (a) Find the uniformly most powerful (UMP) testing procedure of level $\alpha \in (0, 1)$.
- (b) Find the power function $\beta_n(\theta)$ of the UMP test and sketch its graph.
- (c) Show that the UMP test is consistent.
- (d) Construct a finite sample exact $(1 - \alpha)$ confidence interval for θ .

2 Minimal Sufficient Statistics (25 points)

Let a random sample $\mathbf{X}$ with values in the sample space $\mathcal{X}$ have a distribution $\mathbb{P}_\theta$, where $\mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta\}$ is a parametric family dominated by some (σ -finite) measure with strictly positive densities $\{f(\mathbf{x}; \theta) : \theta \in \Theta\}$. To avoid measurability issues, assume $\mathcal{X}$ is finite or countable and all its subsets are measurable. Suppose that there exists a statistic T such that, for every $\mathbf{x}, \mathbf{y} \in \mathcal{X}$, the ratio $f(\mathbf{x}; \theta)/f(\mathbf{y}; \theta)$ does not depend on θ if and only if $T(\mathbf{x}) = T(\mathbf{y})$.

1. Let $T^{-1}(t) = \{\mathbf{x} : T(\mathbf{x}) = t\}$ for each $t \in R_T = \{t : t = T(\mathbf{x}), \mathbf{x} \in \mathcal{X}\}$.

Show that there is a function $R_T \ni t \mapsto \mathbf{y}_t \in \mathcal{X}$ such that for all $\mathbf{x} \in \mathcal{X}$ we have

$$T(\mathbf{x}) = T(\mathbf{y}_{T(\mathbf{x})}).$$

2. Show that T is a sufficient statistic (for $\mathcal{P}$).
3. Let S be another sufficient statistic (for $\mathcal{P}$). Show that, for every $\mathbf{x}, \mathbf{y} \in \mathcal{X}$, the equality $S(\mathbf{x}) = S(\mathbf{y})$ implies that the ratio $f(\mathbf{x}; \theta)/f(\mathbf{y}; \theta)$ does not depend on θ .
4. Show that T is a minimal sufficient statistic (for $\mathcal{P}$).
Hint: $A(\cdot)$ is a function of $B(\cdot)$ if and only if $B(u) = B(v) \implies A(u) = A(v)$ for all u, v .
5. Suppose that $\mathcal{P}$ belongs to the two-dimensional exponential family.
Give sufficient conditions for the existence of a two-dimensional minimal sufficient statistics.

3 MSE Prediction and Identification (25 points)

Let $(Y, \mathbf{X}, \mathbf{W})$ be a random element, where Y has values in $\mathbb{R}$, $\mathbf{X}$ has values in $\mathbb{R}^{d_x}$, and $\mathbf{W}$ has values in $\mathbb{R}^{d_w}$. Assume $\mathbb{E}[Y^2] < \infty$ and $\mathbb{E}[\mathbf{X}'\mathbf{X}] < \infty$. Consider the following MSE prediction problem with respect to the class of partially linear models:

$$(\beta^*, h^*) \in \arg \min_{g \in \mathcal{G}_{\text{PL}}} \mathbb{E}[(Y - g(\mathbf{X}, \mathbf{W}))^2]$$

where the feasibility set is

$$\mathcal{G}_{\text{PL}} = \{g(\mathbf{x}, \mathbf{w}) = \mathbf{x}'\beta + h(\mathbf{w}) : \beta \in \mathbb{R}^{d_x}, h \in \mathcal{H}\}, \quad \mathcal{H} = \{h : \mathbb{E}[|h(\mathbf{W})|^2] < \infty\}.$$

1. Suppose that β^* is known, and find the ($\mathbb{P}_{\mathbf{W}}$ -a.s. unique) solution h^* .
2. Given the solution in the previous part, find the set of optimal solutions for the parameter β .
3. Give sufficient conditions so that $(\beta^*, h^*(\mathbf{W}))$ are (a.s.) unique.
4. Assume that $\mathbb{E}[Y|\mathbf{X}, \mathbf{W}] = \mathbf{X}'\beta_0 + h_0(\mathbf{W})$, with $(\beta_0, h_0(\mathbf{W}))$ are (a.s.) unique.
 - (a) Show that $\beta^* = \beta_0$ and $h^*(\mathbf{W}) = h_0(\mathbf{W})$ (a.s.).
 - (b) Show that

$$\min_{h \in \mathcal{H}} \mathbb{E}[(Y - h(\mathbf{W}))^2] - \min_{g \in \mathcal{G}_{\text{PL}}} \mathbb{E}[(Y - g(\mathbf{X}, \mathbf{W}))^2] = \beta_0' \mathbb{E}[\mathbb{V}[\mathbf{X}|\mathbf{W}]] \beta_0.$$

When is it strictly better (in MSE sense) to include $\mathbf{X}$ when constructing the best MSE predictor of Y given $\mathbf{X}$ and $\mathbf{W}$?

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$